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OF

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A. R. FORSYTH.

19 May, 1897.

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ON CURVILINEAR COORDINATES.

[*From the Quarterly Journal of Pure and Applied Mathematics*, vol. XIX. (1883), pp. 1—22.]

THE present memoir is based upon Mr Warren's "Exercises in Curvilinear and Normal Coordinates," *Camb. Phil. Trans.* t. XII. (1877), pp. 455—502, and has for a principal object the establishment of the six differential equations of the second order corresponding to his six equations for normal coordinates: but the notation is different; the results are more general, inasmuch as I use throughout general curvilinear coordinates instead of his normal coordinates; and as regards my six equations for general curvilinear coordinates, the terms containing differential coefficients of the first order are presented under a different form.

If the position of a point in space is determined by the rectangular coordinates x, y, z; then p, q, r being each of them a given function of x, y, z, we have conversely x, y, z, each of them a given function of p, q, r, which are thus in effect coordinates serving to determine the position of the point, and are called curvilinear coordinates.

But it is not in the first instance necessary to regard x, y, z as rectangular coordinates, or even as Cartesian coordinates at all; we are simply concerned with the two sets of variables x, y, z and p, q, r, each variable of the one set being a given function of the variables of the other set; and, in particular, the x, y, z are regarded as being each of them a given function of the p, q, r.

Except as regards the symbols ξ, η, ζ presently mentioned, the suffixes 1, 2, 3 refer to the variables p, q, r respectively, and are used to denote differentiations in regard to these variables, viz.

$$x_1,\ x_2,\ x_3,\ x_{11},\ x_{22},\ x_{33},\ \ldots$$

are written to denote

$$\frac{dx}{dp},\ \frac{dx}{dq},\ \frac{dx}{dr},\ \frac{d^2x}{dp^2},\ \frac{d^2x}{dq^2},\ \frac{d^2x}{dr^2},\ \&\text{c.},$$

and so in other cases; in particular,

$$\begin{array}{ll} x_1,\ x_2,\ x_3 \text{ denote} & \frac{dx}{dp},\ \frac{dx}{dq},\ \frac{dx}{dr}, \\ y_1,\ y_2,\ y_3 \quad ,, & \frac{dy}{dp},\ \frac{dy}{dq},\ \frac{dy}{dr}, \\ z_1,\ z_2,\ z_3 \quad ,, & \frac{dz}{dp},\ \frac{dz}{dq},\ \frac{dz}{dr}. \end{array}$$

The minors formed with these differential coefficients are denoted by suffixed letters ξ, η, ζ, thus

$$\begin{array}{ll} \xi_1,\ \xi_2,\ \xi_3 \text{ denote} & y_2z_3 - y_3z_2,\ y_3z_1 - y_1z_3,\ y_1z_2 - y_2z_1, \\ \eta_1,\ \eta_2,\ \eta_3 \quad ,, & z_2x_3 - z_3x_2,\ z_3x_1 - z_1x_3,\ z_1x_2 - z_2x_1, \\ \zeta_1,\ \zeta_2,\ \zeta_3 \quad ,, & x_2y_3 - x_3y_2,\ x_3y_1 - x_1y_3,\ x_1y_2 - x_2y_1, \end{array}$$

so that, as regards these letters ξ, η, ζ, the suffixes do not denote differentiations.

The determinant

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \text{ is put} = L;$$

and the symbols (a, b, c, f, g, h), (A, B, C, F, G, H) are defined as follows:

$$\begin{array}{ll} \mathrm{a} = x_1^2 + y_1^2 + z_1^2, & A = \xi_1^2 + \eta_1^2 + \zeta_1^2, \\ \mathrm{b} = x_2^2 + y_2^2 + z_2^2, & B = \xi_2^2 + \eta_2^2 + \zeta_2^2, \\ \mathrm{c} = x_3^2 + y_3^2 + z_3^2, & C = \xi_3^2 + \eta_3^2 + \zeta_3^2, \\ \mathrm{f} = x_2x_3 + y_2y_3 + z_2z_3, & F = \xi_2\xi_3 + \eta_2\eta_3 + \zeta_2\zeta_3, \\ \mathrm{g} = x_3x_1 + y_3y_1 + z_3z_1, & G = \xi_3\xi_1 + \eta_3\eta_1 + \zeta_3\zeta_1, \\ \mathrm{h} = x_1x_2 + y_1y_2 + z_1z_2, & H = \xi_1\xi_2 + \eta_1\eta_2 + \zeta_1\zeta_2. \end{array}$$

We have then, further,

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = L, \qquad \begin{vmatrix} \mathrm{a} & \mathrm{h} & \mathrm{g} \\ \mathrm{h} & \mathrm{b} & \mathrm{f} \\ \mathrm{g} & \mathrm{f} & \mathrm{c} \end{vmatrix} = L^2,$$

$$\begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix} = L^2, \qquad \begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = \frac{1}{L^2};$$

$$(A,\ B,\ C,\ F,\ G,\ H) = (\mathrm{bc} - \mathrm{f}^2,\ \mathrm{ca} - \mathrm{g}^2,\ \mathrm{ab} - \mathrm{h}^2,\ \mathrm{gh} - \mathrm{af},\ \mathrm{hf} - \mathrm{bg},\ \mathrm{fg} - \mathrm{ch}),$$

$$L^2\,(\mathrm{a},\ \mathrm{b},\ \mathrm{c},\ \mathrm{f},\ \mathrm{g},\ \mathrm{h}) = (BC{-}F^2,\ CA{-}G^2,\ AB{-}H^2,\ GH{-}AF,\ HF{-}BG,\ FG{-}CH),$$

which equations are at once proved, and are fundamental ones in the theory.

It is convenient to add that we have

$$\begin{vmatrix} \dfrac{dp}{dx} & \dfrac{dp}{dy} & \dfrac{dp}{dz} \\ \dfrac{dq}{dx} & \dfrac{dq}{dy} & \dfrac{dq}{dz} \\ \dfrac{dr}{dx} & \dfrac{dr}{dy} & \dfrac{dr}{dz} \end{vmatrix} = \frac{1}{L} \begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix},$$

that is, $\dfrac{dp}{dx} = \dfrac{1}{L}\,\xi_1$, &c.; and, further,

$$dx^2 + dy^2 + dz^2 = (\mathrm{a},\ \mathrm{b},\ \mathrm{c},\ \mathrm{f},\ \mathrm{g},\ \mathrm{h} \mid dp,\ dq,\ dr)^2,$$

$$dp^2 + dq^2 + dr^2 = \frac{1}{L^2}\,(A,\ B,\ C,\ F,\ G,\ H \mid dx,\ dy,\ dz)^2.$$

Differentiating the values of a, b, c, f, g, h, we have $\frac{1}{2}\mathrm{a}_1 = x_1x_{11} + y_1y_{11} + z_1z_{11}$, which may be written in the abbreviated form $\frac{1}{2}\mathrm{a}_1 = 1.11$; similarly

$$x_1x_{12} + x_2x_{11} + y_1y_{12} + y_2y_{11} + z_1z_{12} + z_2z_{11},$$

which in like manner may be written $\mathrm{f}_1 = 2.13 + 3.12$, and so in other cases; observe that, in the duad part of any symbol, the order of the numbers is immaterial, $2.13 = 2.31$. The whole system of equations is

$$\begin{array}{lll}
\frac{1}{2}\mathrm{a}_1 = 1.11, & \frac{1}{2}\mathrm{a}_2 = 1.12, & \frac{1}{2}\mathrm{a}_3 = 1.13, \\
\frac{1}{2}\mathrm{b}_1 = 2.12, & \frac{1}{2}\mathrm{b}_2 = 2.22, & \frac{1}{2}\mathrm{b}_3 = 2.23, \\
\frac{1}{2}\mathrm{c}_1 = 3.13, & \frac{1}{2}\mathrm{c}_2 = 3.23, & \frac{1}{2}\mathrm{c}_3 = 3.33, \\
\mathrm{f}_1 = 2.13 + 3.12, & \mathrm{f}_2 = 2.23 + 3.22, & \mathrm{f}_3 = 2.33 + 3.23, \\
\mathrm{g}_1 = 3.11 + 1.13, & \mathrm{g}_2 = 3.12 + 1.23, & \mathrm{g}_3 = 3.13 + 1.33, \\
\mathrm{h}_1 = 1.12 + 2.11, & \mathrm{h}_2 = 1.22 + 2.12, & \mathrm{h}_3 = 1.23 + 2.13.
\end{array}$$

These may also be written

$$\begin{array}{lll}
1.11 = \frac{1}{2}\mathrm{a}_1, & 1.22 = \mathrm{h}_2 - \frac{1}{2}\mathrm{b}_1, & 1.33 = \mathrm{g}_3 - \frac{1}{2}\mathrm{c}_1, \\
2.11 = \mathrm{h}_1 - \frac{1}{2}\mathrm{a}_2, & 2.22 = \frac{1}{2}\mathrm{b}_2, & 2.33 = \mathrm{f}_3 - \frac{1}{2}\mathrm{c}_2, \\
3.11 = \mathrm{g}_1 - \frac{1}{2}\mathrm{a}_3, & 3.22 = \mathrm{f}_2 - \frac{1}{2}\mathrm{b}_3, & 3.33 = \frac{1}{2}\mathrm{c}_3, \\
1.23 = \frac{1}{2}(-\mathrm{f}_1 + \mathrm{g}_2 + \mathrm{h}_3), & 1.31 = \frac{1}{2}\mathrm{a}_3, & 1.12 = \frac{1}{2}\mathrm{a}_2, \\
2.23 = \frac{1}{2}\mathrm{b}_3, & 2.31 = \frac{1}{2}(\mathrm{f}_1 - \mathrm{g}_2 + \mathrm{h}_3), & 2.12 = \frac{1}{2}\mathrm{b}_1, \\
3.23 = \frac{1}{2}\mathrm{c}_2, & 3.31 = \frac{1}{2}\mathrm{c}_1, & 3.12 = \frac{1}{2}(\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3).
\end{array}$$

It is to be observed that we can, from each system of three equations, express a set of second differential coefficients of the x, y, z in terms of the first differential

1—2

coefficients of the a, b, c, f, g, h; thus the three equations containing 11, written at length, are

$$\begin{aligned} x_1 x_{11} + y_1 y_{11} + z_1 z_{11} &= \tfrac{1}{2}\mathrm{a}_1, \\ x_2 \quad + y_2 \quad + z_2 \quad &= \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2, \\ x_3 \quad + y_3 \quad + z_3 \quad &= \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3, \end{aligned}$$

three linear equations for the determination of x_{11}, y_{11}, z_{11}; hence for x_{11} we have

$$x_{11}\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \tfrac{1}{2}\mathrm{a}_1(y_2z_3-y_3z_2)+(\mathrm{h}_1-\tfrac{1}{2}\mathrm{a}_2)(y_3z_1-y_1z_3)+(\mathrm{g}_1-\tfrac{1}{2}\mathrm{a}_3)(y_1z_2-y_2z_1),$$

or, what is the same thing,

$$L\,x_{11} = \tfrac{1}{2}\mathrm{a}_1\,\xi_1 + (\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2)\,\xi_2 + (\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)\,\xi_3,$$

and so for y_{11}, z_{11}; or if (as in the sequel) we desire the value of a linear function $\alpha x_{11} + \beta y_{11} + \gamma z_{11}$, calling this for a moment $\square$, we join to the foregoing a new equation

$$\alpha x_1 + \beta y_1 + \gamma z_1 = \square,$$

and then, eliminating the three quantities, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \square \\ x_1 & y_1 & z_1 & \tfrac{1}{2}\mathrm{a}_1 \\ x_2 & y_2 & z_2 & \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2 \\ x_3 & y_3 & z_3 & \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3 \end{vmatrix} = 0,$$

giving $L\,\square$ as a linear function of $\tfrac{1}{2}\mathrm{a}_1$, $\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2$, $\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3$.

We can in like manner form the expressions for the second differential coefficients of the a, b, c, f, g, h: these will of course contain third differential coefficients of the x, y, z.

Writing down only what is wanted, we have

$$\begin{aligned} \tfrac{1}{2}\mathrm{a}_{22} &= 12.12 + 1.122, \\ \tfrac{1}{2}\mathrm{b}_{11} &= 12.12 + 2.112, \\ \mathrm{h}_{12} &= 12.12 + 11.22 + 1.122 + 2.112, \end{aligned}$$

where of course 12.12 denotes $x_{12}^2+y_{12}^2+z_{12}^2$, 1.122 denotes $x_1.x_{122}+y_1.y_{122}+z_1.z_{122}$, and so in other cases: it follows that

$$\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) = 12.12 - 11.22,$$

so that the third differential coefficients of x, y, z, which enter into the expression of $\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}$ destroy each other, and this combination contains really only second differential coefficients of x, y, z.

Similarly

$$
\begin{aligned}
\mathrm{f}_{12} &= 31.23 + 33.12 + 3.123 + 2.133,\\
\mathrm{g}_{31} &= 31.23 + 33.12 + 3.123 + 1.233,\\
\mathrm{c}_{12} &= 2 \cdot 31.23 + 2 \cdot 3.123,\\
\mathrm{h}_{33} &= 2 \cdot 31.23 + 1.233 + 2.133,
\end{aligned}
$$

and thence

$$\mathrm{f}_{12} + \mathrm{g}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33} = -2\,(31.23 - 33.12),$$

so that here again we have a combination containing only second differential coefficients of x, y, z.

There are thus, in all, the six combinations

$$
\begin{aligned}
\mathrm{b}_{33} + \mathrm{c}_{22} - 2\mathrm{f}_{23} &\quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (\mathfrak{A}),\\
\mathrm{c}_{11} + \mathrm{a}_{33} - 2\mathrm{g}_{31} &\quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (\mathfrak{B}),\\
\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12} &\quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (\mathfrak{C}),\\
\mathrm{g}_{23} + \mathrm{h}_{32} - \mathrm{a}_{23} - \mathrm{f}_{11} &\quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (\mathfrak{f}),\\
\mathrm{h}_{31} + \mathrm{f}_{13} - \mathrm{b}_{31} - \mathrm{g}_{22} &\quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (\mathfrak{g}),\\
\mathrm{f}_{12} + \mathrm{g}_{21} - \mathrm{c}_{12} - \mathrm{h}_{33} &\quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (\mathfrak{h}),
\end{aligned}
$$

each really containing only second differential coefficients of x, y, z; and we thus understand how each of these combinations may be expressible in terms of the first differential coefficients of $(\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{f}, \mathrm{g}, \mathrm{h})$. We have, in fact, for thus expressing these combinations the six equations called $(\mathfrak{A})$, $(\mathfrak{B})$, $(\mathfrak{C})$, $(\mathfrak{f})$, $(\mathfrak{g})$, $(\mathfrak{h})$ about to be obtained, and which are the generalisations of Warren's six equations for normal coordinates.

I consider the several determinants of the form

$$\begin{vmatrix} x_{\alpha\beta} & y_{\alpha\beta} & z_{\alpha\beta} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix},$$

in all 18, since the suffixes for the top row may be 11, 22, 33, 23, 31, 12, and those for the second and third rows 2, 3; 3, 1; or 1, 2. Each determinant is a linear function of second differential coefficients of x, y, z (thus the determinant written down is $\xi_1 x_{11} + \eta_1 y_{11} + \zeta_1 z_{11}$, and as such it can, by what precedes, be expressed by means of the first differential coefficients of the a, b, c, f, g, h. Thus, if the determinant above written down be called D, writing ξ_3, η_3, ζ_3, for α, β, γ, we have

$$\begin{vmatrix} \xi_3 & \eta_3 & \zeta_3 & \square \\ x_1 & y_1 & z_1 & \frac{1}{2}\mathrm{a}_1 \\ x_2 & y_2 & z_2 & \mathrm{h}_1 - \frac{1}{2}\mathrm{a}_2 \\ x_3 & y_3 & z_3 & \mathrm{g}_1 - \frac{1}{2}\mathrm{a}_3 \end{vmatrix} = 0,$$

that is,

$$L\,\square = -\begin{vmatrix} & \xi_3 & \eta_3 & \zeta_3 \\ \frac{1}{2}\mathrm{a}_1 & x_1 & y_1 & z_1 \\ \mathrm{h}_1 - \frac{1}{2}\mathrm{a}_2 & x_2 & y_2 & z_2 \\ \mathrm{g}_1 - \frac{1}{2}\mathrm{a}_3 & x_3 & y_3 & z_3 \end{vmatrix},$$